

Google Cloud Vertex AI User Manual

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Google Cloud offers a managed machine learning platform called Vertex AI. Vertex AI enables developers to build, train, evaluate, and deploy machine learning models. Vertex AI supports Google's Gemini family of models. Google Cloud developed Vertex AI to simplify AI application development. Organizations can deploy Gemini models through Vertex AI without managing infrastructure.

Gemini Models

Gemini 1.5 Flash

Gemini 1.5 Pro

Gemini 2.5 Flash

Gemini 2.5 Pro

Gemini 2.5 Pro provides improved reasoning capabilities. Gemini 2.5 Flash is optimized for low latency.

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Model Deployment

A model must first be uploaded to Vertex AI. After uploading, an Endpoint can be created. The Endpoint hosts one or more deployed models. Applications communicate with deployed models through REST APIs. Each endpoint has a unique Endpoint ID. Traffic can be split across multiple deployed models.

Authentication

Google Cloud uses IAM for authentication. Applications authenticate using Service Accounts. A Service Account contains credentials used to access Vertex AI. IAM assigns roles and permissions. A Service Account must have the Vertex AI User role before invoking a model.

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Storage

Vertex AI stores datasets inside Google Cloud Storage. Training jobs read datasets from Cloud Storage buckets. Models are stored inside the Vertex AI Model Registry. The Model Registry maintains model versions. Each model version contains metadata.

Training

A Training Pipeline executes multiple steps. Datasets are preprocessed before training. The preprocessing step removes duplicate records. Training produces a trained model. Evaluation calculates accuracy, precision, and recall. Only approved models are registered.

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Monitoring

Vertex AI continuously monitors deployed models. Monitoring detects model drift. Monitoring also detects feature drift. Alerts are generated whenever drift exceeds configured thresholds. Cloud Monitoring receives these alerts.

Logging

Cloud Logging stores inference logs. Each inference request generates a log entry. Logs include timestamp, endpoint ID, model version, latency, and response status. Logs help administrators troubleshoot production issues.

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Networking

Vertex AI endpoints can be private or public. Private endpoints require VPC connectivity. Public endpoints require authentication. Load Balancers distribute incoming requests. Traffic first reaches the Load Balancer before being routed to an Endpoint.

Security

Vertex AI encrypts stored data. Google Cloud manages encryption keys by default. Customers may configure Customer Managed Encryption Keys (CMEK). IAM policies control resource access. Audit Logs record administrative activities.

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Relationships Summary

Google Cloud developed Vertex AI.
Vertex AI supports Gemini 2.5 Pro.
Vertex AI supports Gemini 2.5 Flash.
Vertex AI uses Cloud Storage.
Training Pipeline preprocesses Dataset.
Training Pipeline produces Model.
Model Registry stores Model.
Model is deployed to Endpoint.
Endpoint receives Requests.
Cloud Logging stores Logs.
Cloud Monitoring receives Alerts.
Service Account authenticates Application.
IAM grants Roles.
Roles authorize Service Accounts.
Load Balancer routes Requests to Endpoint.
Private Endpoint uses VPC.